

Mitul Dhawan

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EXPERIENCE

Software Engineer

Sept 2025 – July 2026

Eden

Remote

- **Developed FastTelemetry** (open-source Rust), a high-performance OpenTelemetry alternative using sharded thread-local counters, cutting cache contention on the hot recording path (**2–3x faster** counters, up to **40x** under contention).
- **Architected a 94+ metric OpenTelemetry system in Rust** across ~10 categories (core, LLM, IAM, proxy, migration, load balancer, and more), exported to **Datadog**/Grafana via OTLP.
- **Built a custom Rust logging framework 3x faster than env_logger** and a **Criterion**-benchmarked metrics batching system, minimizing observability overhead on critical paths.
- Conducted **code reviews**, resolved critical bug fixes, and wrote documentation that shortened onboarding for new engineers.

Co-Founder & Backend Engineer

Apr 2025 – May 2026

ZeroBlock

Remote, Part-time

- **Co-founded and architected ZeroBlock**, an ultra-fast Solana trading extension for Axiom.trade, building the **entire backend from scratch in Rust + Actix Web** for high-throughput, low-latency execution.
- **Designed a distributed data architecture: ScyllaDB** for persistence and **Valkey/Redis** for real-time caching and session state, sustaining thousands of concurrent requests.
- **Owned infra: Solana on-chain programs**, a market maker, a shredstream proxy, and **Kubernetes** deployment with CI/CD and in-house TLS cert automation.

Software Engineer

Dec 2023 – Jul 2025

Bitwyre

Remote

- **Built and optimized low-latency C++ trading systems** (matching and risk engines) for a live exchange, shipping performance and correctness improvements, and later **porting them to Rust**.
- **Built market-making systems** in C++ and Python placing orders within milliseconds, consuming live market data over **WebSockets** and **Apache Kafka**.
- **Promoted twice in 18 months** across software and quantitative teams; triaged production incidents with root-cause analysis.

PROJECTS

ZeroLag • Rust, DPDK, custom TCP/IP, Linux

2026

- **Low-latency message queue** with a kernel-bypass networking stack (DPDK) and a built-in observability suite; a systems project in the class of Kafka and NATS.

Emulators (8080 & NES) • Rust, WebAssembly, SDL2

2026

- **CPUs written from scratch in Rust**: a cycle-accurate Intel 8080 (WASM + native, live in-browser) and an in-progress NES emulator with a modular core/frontend workspace.

TECHNICAL SKILLS

Languages: Rust, C++, Python, Go, TypeScript/JavaScript, SQL

Rust Stack: Tokio, Actix Web, Tonic (gRPC), Serde, sonic-rs, Tracing; proc macros; open-source Rust (FastTelemetry)

Systems & Performance: performance profiling and optimization, concurrency and multi-threading, low-latency systems, reliability, observability

Distributed & Infra: distributed and event-driven systems (Kafka, WebSockets), gRPC, Kubernetes, Docker, Linux; ScyllaDB, Redis, PostgreSQL

Observability: Datadog, OpenTelemetry, Prometheus, Grafana; metrics, tracing, incident triage

AI & Craftsmanship: AI-assisted dev (Claude Code, Cursor), LLM/MCP; testing, documentation, clean modular APIs

EDUCATION

Southern Alberta Institute of Technology

Calgary, AB

Diploma in Software Development, GPA: 3.94/4.0

Dec 2023 – Aug 2024